

EXPERIMENT 17

High Level Programming Extensions – Functions

Aim:

To implement Functions using program in MySQL.

Functions:

A function is a subprogram that computes a value.

Creating a function

The CREATE FUNCTION statement is also used in MySQL to support UDFs (user-defined functions). A UDF can be regarded as an external stored function.

MySQL stored function syntax

```
CREATE FUNCTION function_name(param1,param2,...)
```

```
RETURNS datatype
```

```
[NOT] DETERMINISTIC
```

```
.....
```

```
Statements
```

```
....
```

Questions

1. Create a MySQL function GetPatientLevel() that accepts a patient's age and returns:

- Age > 60 → SENIOR
- Age between 18 and 60 → ADULT
- Age < 18 → CHILD

2. Write a recursive MySQL procedure to calculate the factorial of a number.

OUTPUT:

1)

```
3 • DROP FUNCTION IF EXISTS GetPatientLevel;
4
5 DELIMITER //
6
7 • CREATE FUNCTION GetPatientLevel(patient_age INT)
8 RETURNS VARCHAR(20)
9 DETERMINISTIC
10 BEGIN
11     DECLARE patient_level VARCHAR(20);
12
13     IF patient_age > 60 THEN
14         SET patient_level = 'SENIOR';
15
16     ELSEIF patient_age >= 18 THEN
17         SET patient_level = 'ADULT';
18
19     ELSE
20         SET patient_level = 'CHILD';
21     END IF;
22
23     RETURN patient_level;
24 END //
25
26 DELIMITER ;
27
28 • SELECT GetPatientLevel(25) AS Patient_Level;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Patient_Level			
▶	ADULT			

2)

```
31
32 • SET GLOBAL max_sp_recursion_depth = 10;
33
34 • SET SESSION max_sp_recursion_depth = 10;
35
36 • DROP PROCEDURE IF EXISTS Factorial;
37
38 DELIMITER //
39
40 • CREATE PROCEDURE Factorial(
41     IN n INT,
42     OUT result INT
43 )
44 BEGIN
45     DECLARE temp INT;
46
47     IF n = 0 OR n = 1 THEN
48         SET result = 1;
49     ELSE
50         CALL Factorial(n - 1, temp);
51         SET result = n * temp;
52     END IF;
53 END //
54
```

Result Grid		Filter Rows:	Export:	Wrap Cell Con
	Factorial_Result			
▶	120			

RESULT:

Thus the Functions using program in MySQL executed successfully